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SUBJECT : DSA0604 - DATA HANDLING AND VISUALISATION
LAB EXPERIMENTS - 16 TO 20

Set 21

Dataset: Energy_Consumption_Data.csv

Sector Region Month Temperature
(°C)
Units_Consumed
(kWh)

Cost
(₹)

Renewable_Usage
(%) Peak_Hours

Residential North Jan 15 320 2100 22 4
Commercial South Jan 24 540 3600 18 6
Industrial West Feb 20 880 5900 12 8
Residential East Feb 18 350 2300 25 5
Commercial North Mar 28 610 4100 20 7
Industrial South Mar 30 920 6200 15 9

1. Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.
2. Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.
3. Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.
4. Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

CODE AND OUTPUTS:

>HISTOGRAM

```
#=====
```

```
# Question 1
```

```
# Histogram and Density Plot
```

```
#=====
```

```
energy <- read.csv("C:/Users/selco/OneDrive/Desktop/Energy_Consumption_Data.csv")
```

```
print(energy)
```

```
hist(energy$Units_Consumed,  
     probability = TRUE,  
     col = "lightblue",  
     border = "black",  
     main = "Histogram of Units Consumed",  
     xlab = "Units Consumed (kWh)")
```

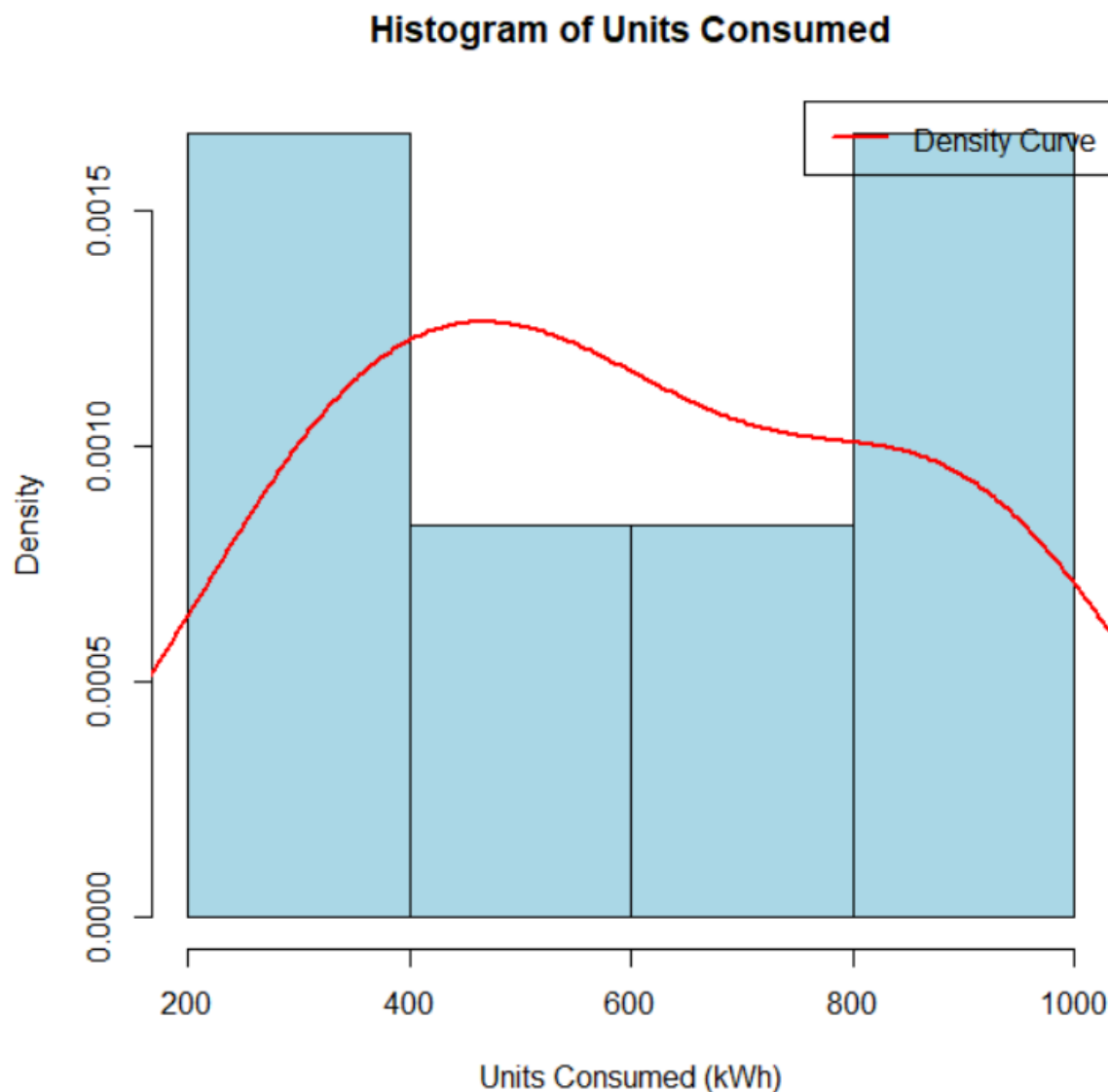
```
lines(density(energy$Units_Consumed),  
      col = "red",  
      lwd = 2)
```

```
legend("topright",  
      legend = "Density Curve",  
      col = "red",  
      lwd = 2)
```

VALUES IN CSV

```
> print(energy)
```

	Sector	Region	Month	Temperature	Units_Consumed	Cost	Renewable_Usage
1	Residential	North	Jan	15	320	2100	22
2	Commercial	South	Jan	24	540	3600	18
3	Industrial	West	Feb	20	880	5900	12
4	Residential	East	Feb	18	350	2300	25
5	Commercial	North	Mar	28	610	4100	20
6	Industrial	South	Mar	30	920	6200	15
	Peak_Hours						
1					4		
2					6		
3					8		
4					5		
5					7		
6					9		



>Scatter Plot with Bubble Size

```
#=====
```

```
# Question 2
```

```
# Scatter Plot
```

```
#=====
```

```
energy <- read.csv("C:/Users/selco/OneDrive/Desktop/Energy_Consumption_Data.csv")
```

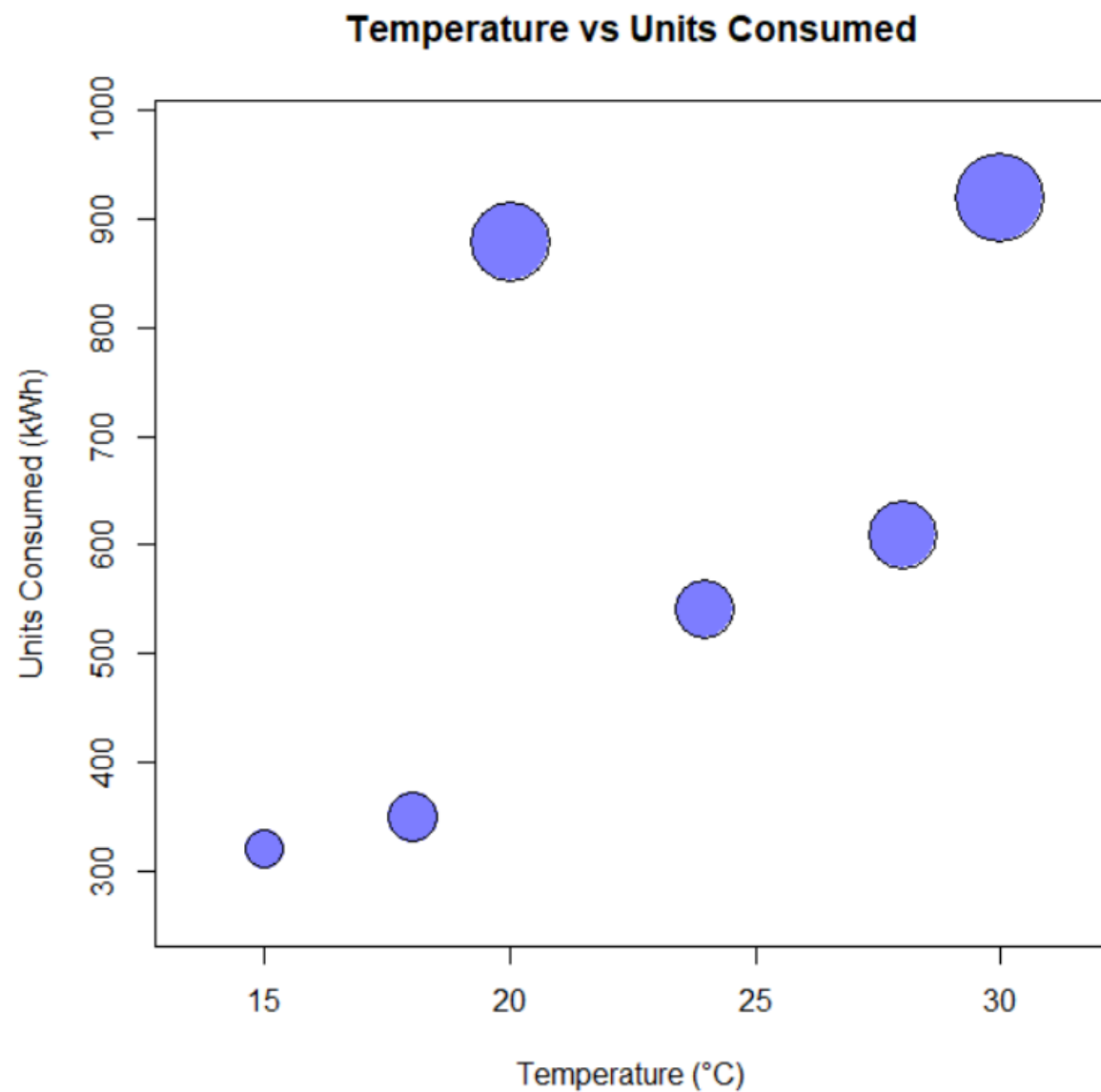
```
print(energy)
```

```
symbols(energy$Temperature,
        energy$Units_Consumed,
        circles = energy$Peak_Hours,
        inches = 0.25,
        bg = rgb(0,0,1,0.5),
        fg = "black",
        xlab = "Temperature (°C)",
```

```
ylab = "Units Consumed (kWh)",  
main = "Temperature vs Units Consumed")
```

VALUES IN CSV

```
> print(energy)  
  Sector Region Month Temperature Units_Consumed Cost Renewable_Usage  
1 Residential North Jan         15          320 2100             22  
2 Commercial South Jan         24          540 3600             18  
3 Industrial West  Feb         20          880 5900             12  
4 Residential East  Feb         18          350 2300             25  
5 Commercial North Mar         28          610 4100             20  
6 Industrial South Mar         30          920 6200             15  
Peak_Hours  
1          4  
2          6  
3          8  
4          5  
5          7  
6          9  
>
```



>BAR CHART- Average Renewable Usage by Sector

```
#=====
```

```
# Question 3
```

```
# Bar Chart
```

```
#=====
```

```
energy <- read.csv("C:/Users/selco/OneDrive/Desktop/Energy_Consumption_Data.csv")
```

```
average <- aggregate(Renewable_Usage ~ Sector,  
                      data = energy,  
                      mean)
```

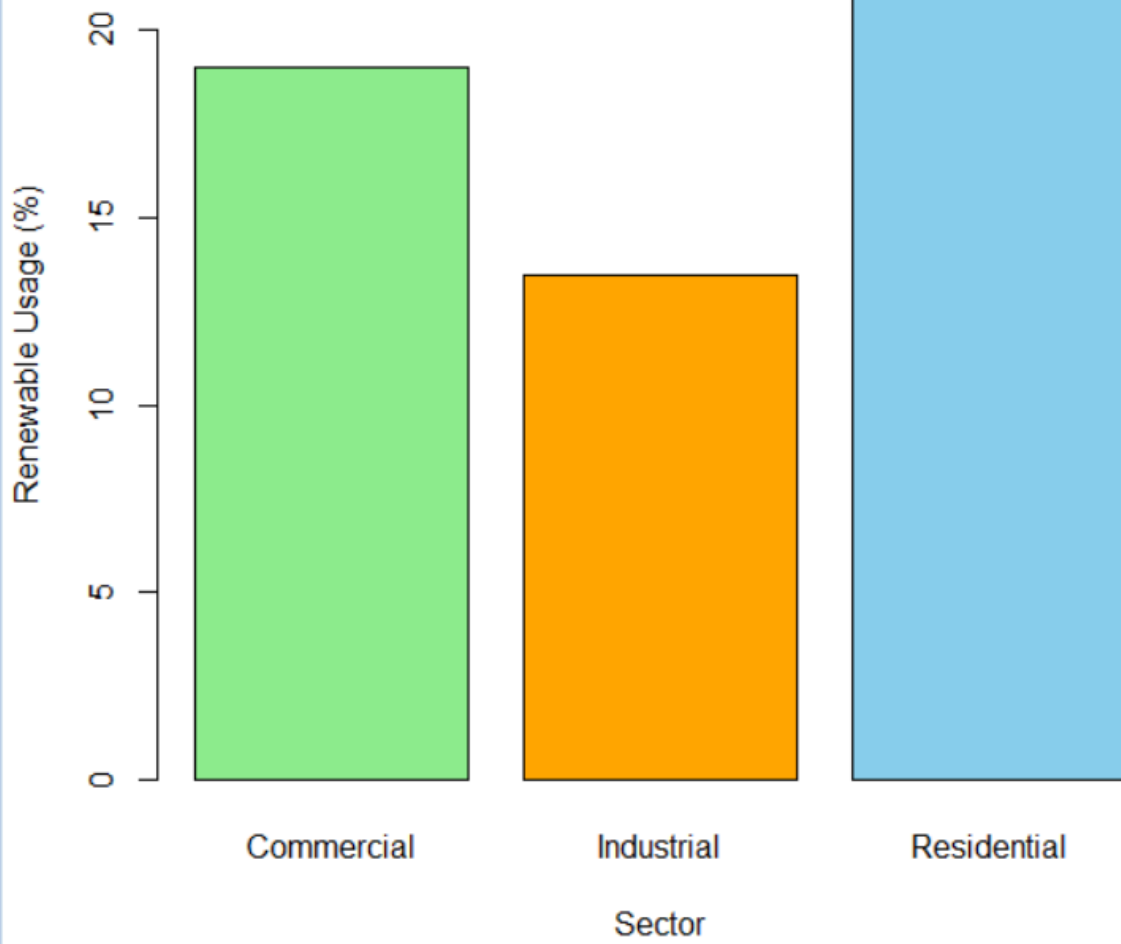
```
print(average)
```

```
barplot(average$Renewable_Usage,  
        names.arg = average$Sector,  
        col = c("lightgreen","orange","skyblue"),  
        main = "Average Renewable Usage by Sector",  
        xlab = "Sector",  
        ylab = "Renewable Usage (%)")
```

VALUES

```
> print(average)  
      Sector Renewable_Usage  
1 Commercial           19.0  
2 Industrial           13.5  
3 Residential          23.5  
>
```

Average Renewable Usage by Sector



Set 22: Time Series Analysis

Dataset: Monthly Sales Data

Month Sales (in \$)

January 15000

February 18000

March 22000

April 20000

May 23000

1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.
2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.
3. In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.
4. In R, create an autocorrelation plot to identify seasonality in the time series data.

CODE AND OUTPUT:

Time Series Line Chart (R)

```
#=====
```

```
# Question 1
```

```
# Time Series Line Chart
```

```
#=====
```

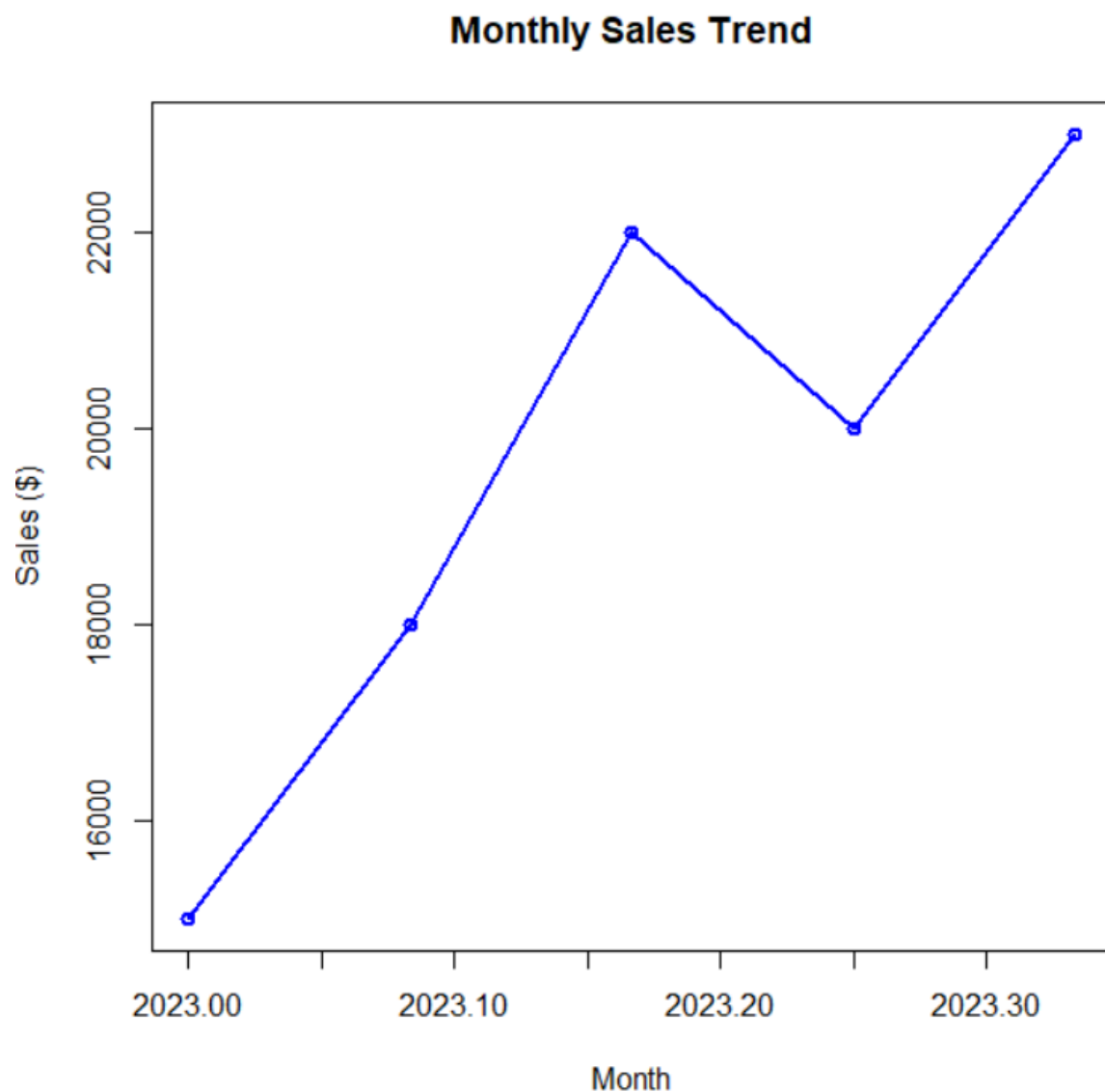
```
sales <- read.csv("C:/Users/selco/OneDrive/Desktop/Monthly_Sales_Data.csv")
```

```
print(sales)
```

```
sales_ts <- ts(sales$Sales, start = c(2023,1), frequency = 12)
```

```
plot(sales_ts,  
     type = "o",  
     col = "blue",  
     lwd = 2,  
     xlab = "Month",  
     ylab = "Sales ($)",  
     main = "Monthly Sales Trend")
```

GRAPH



VALUES

```
>
> print(sales)
      Month Sales Advertising_Budget Product
1 January 15000             3000 Product A
2 February 18000             3500 Product B
3 March 22000             4200 Product A
4 April 20000             3900 Product C
5 May 23000             4500 Product B
>
> # Create a scatter plot of Sales vs Advertising_Budget
> # Add a regression line to the scatter plot
```

>Scatter Plot (Advertising Budget vs Sales)

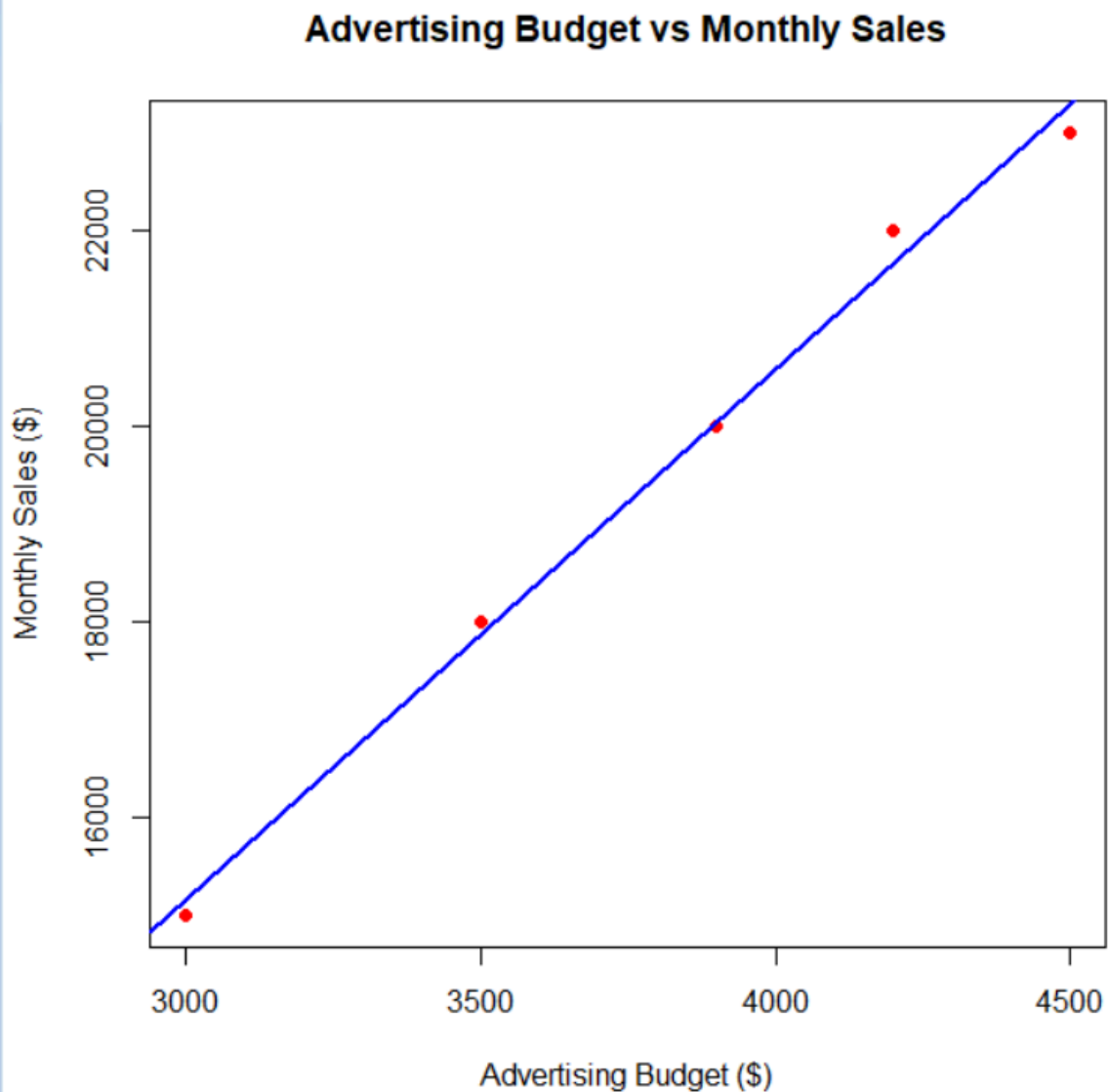
```
#=====
# Question 2
# Scatter Plot
#=====
```

```
sales <- read.csv("C:/Users/selco/OneDrive/Desktop/Monthly_Sales_Data.csv")
```

```
print(sales)
```

```
plot(sales$Advertising_Budget,  
     sales$Sales,  
     pch = 19,  
     col = "red",  
     xlab = "Advertising Budget ($)",  
     ylab = "Monthly Sales ($)",  
     main = "Advertising Budget vs Monthly Sales")
```

```
abline(lm(Sales ~ Advertising_Budget, data = sales),  
       col = "blue",  
       lwd = 2)
```



Set 23: Employee Performance Analysis

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

1 Sales 5 85

2 HR 3 92

3 Marketing 7 78

4 Sales 4 90

5 HR 2 76

1. In R, create a bar chart to visualize the distribution of employees across different departments.

Label the chart elements.

2. In R, generate a line chart to visualize the performance trend of employees over time.

Label the

axes and title the chart.

3. In Tableau, build a dashboard combining a bar chart showing department distribution and a

scatter plot displaying the relationship between years of service and performance scores.

4. In R, develop a table showing the performance data for each employee. Label the table elements.

CODE:

Bar Chart (R)

```
#=====
```

```
# Question 1
```

```
# Bar Chart - Department Distribution
```

```
#=====
```

```
employee <- read.csv("C:/Users/selco/OneDrive/Desktop/Employee_Performance.csv")
```

```
print(employee)
```

```
dept <- table(employee$Department)
```

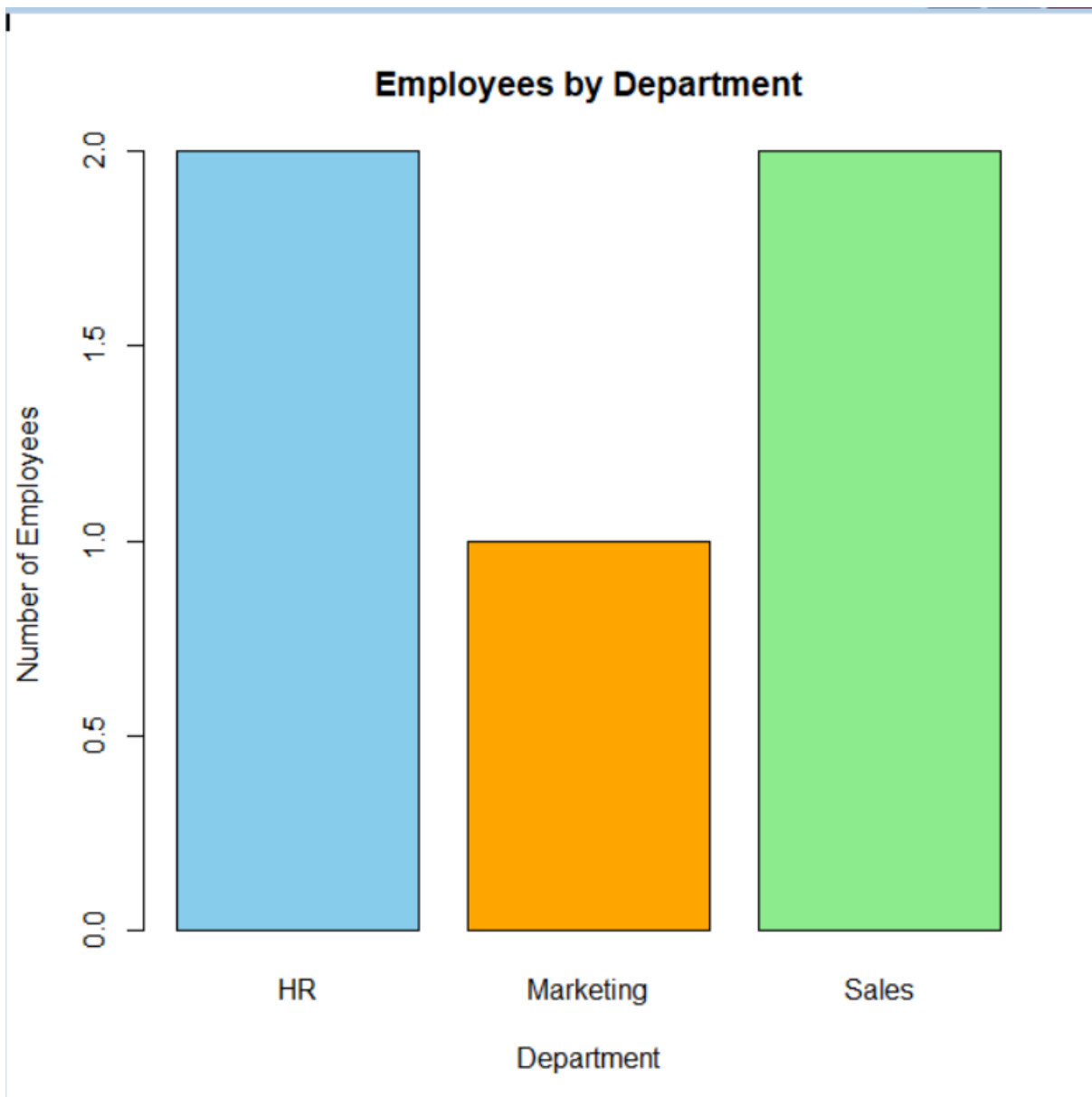
```
barplot(dept,
```

```
  col = c("skyblue","orange","lightgreen"),
```

```
  main = "Employees by Department",
```

```
  xlab = "Department",
```

```
  ylab = "Number of Employees")
```



>Line Chart (R)

#=====

Question 2

Employee Performance Trend

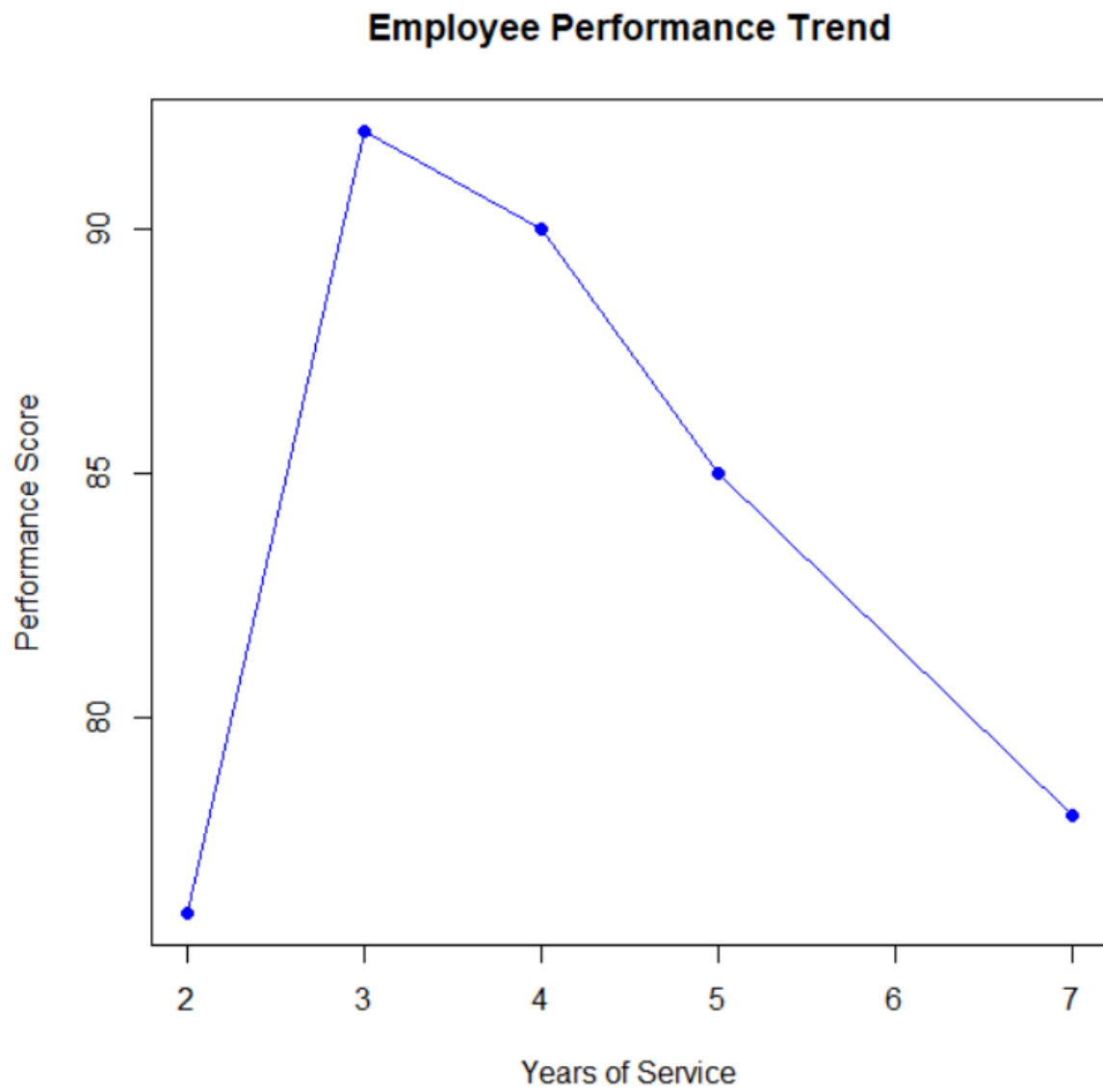
#=====

```
employee <- read.csv("C:/Users/selco/OneDrive/Desktop/Employee_Performance.csv")
```

```
employee <- employee[order(employee$Years_of_Service), ]
```

```
plot(employee$Years_of_Service,  
      employee$Performance_Score,  
      type = "o",  
      pch = 16,  
      col = "blue",  
      xlab = "Years of Service",
```

```
ylab = "Performance Score",  
main = "Employee Performance Trend")
```



Set 24 Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
------------	--------------	--------------------

1	Product A	250
---	-----------	-----

2	Product B	175
---	-----------	-----

3	Product C	300
---	-----------	-----

4	Product D	200
---	-----------	-----

5	Product E	220
---	-----------	-----

1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.

3. In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.

4. In R, create a scatter plot to explore the relationship between product price and quantity available.

Explain any insights.

CODE:

>Bar Chart (R)

```
#=====
```

```
# Question 1
```

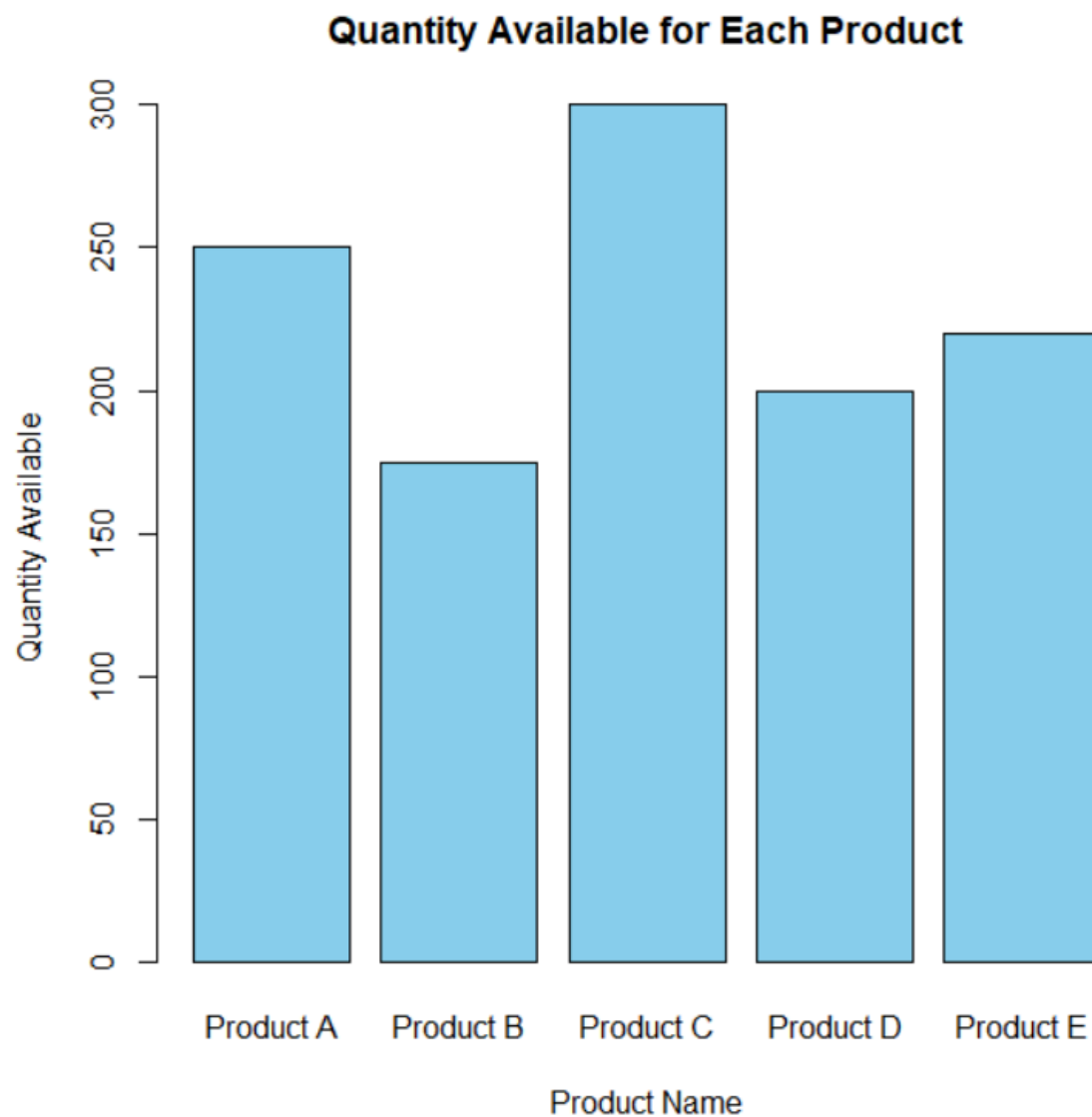
```
# Product Quantity Bar Chart
```

```
#=====
```

```
inventory <- read.csv("C:/Users/selco/OneDrive/Desktop/Product_Inventory.csv")
```

```
print(inventory)
```

```
barplot(inventory$Quantity_Available,  
        names.arg = inventory$Product_Name,  
        col = "skyblue",  
        main = "Quantity Available for Each Product",  
        xlab = "Product Name",  
        ylab = "Quantity Available")
```



>Stacked Bar Chart (R)

```
#=====
# Question 2
# Stacked Bar Chart
#=====
```

```
inventory <- read.csv("C:/Users/selco/OneDrive/Desktop/Product_Inventory.csv")
```

```
library(reshape2)
```

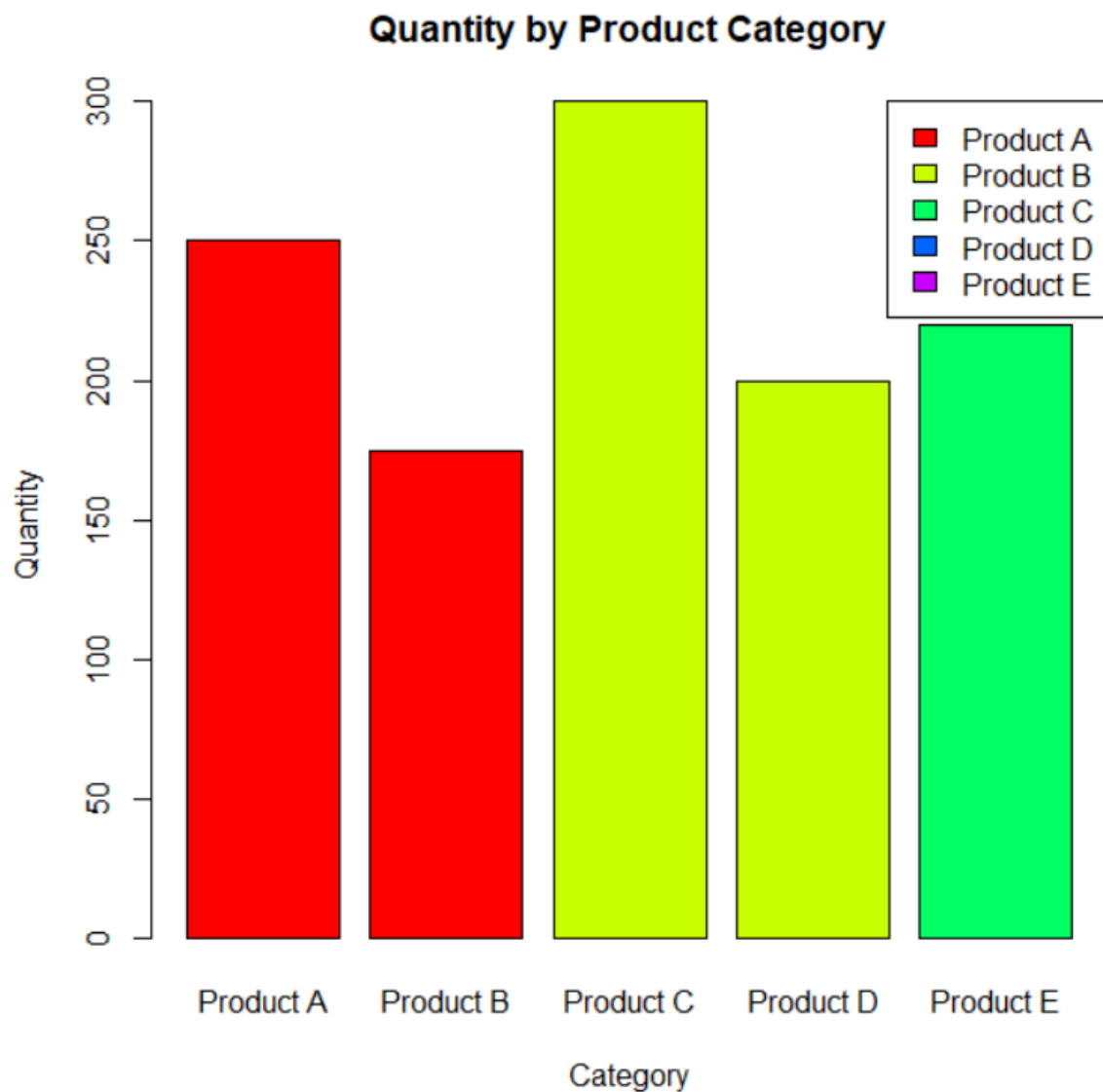
```
inventory_matrix <- dcast(inventory,
  Category~Product_Name,
  value.var="Quantity_Available",
  fill=0)
```

```
rownames(inventory_matrix) <- inventory_matrix$Category
```

```
inventory_matrix <- as.matrix(inventory_matrix[,-1])
```

```
barplot(inventory_matrix,  
  beside = FALSE,  
  col = rainbow(ncol(inventory_matrix)),  
  main = "Quantity by Product Category",  
  xlab = "Category",  
  ylab = "Quantity")
```

```
legend("topright",  
  legend = colnames(inventory_matrix),  
  fill = rainbow(ncol(inventory_matrix)))
```



Set 25: Website Traffic Analysis

Dataset: Website Traffic

Date Page Views Click-through Rate

2023-01-01 1500 2.3%

2023-01-02 1600 2.7%

2023-01-03 1400 2.0%

2023-01-04 1650 2.4%

2023-01-05 1800 2.6%

1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and

title the chart.

2. In R, generate a bar chart to show the top N days with the highest click-through rates.

Label the

chart elements.

3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart

displaying page views by source.

CODE AND OUTPUTS:

R (Bar Chart)

```
#=====
```

```
# Question 2
```

```
# Top N Days with Highest Click Through Rate
```

```
#=====
```

```
traffic <- read.csv("C:/Users/selco/OneDrive/Desktop/Website_Traffic.csv")
```

```
print(traffic)
```

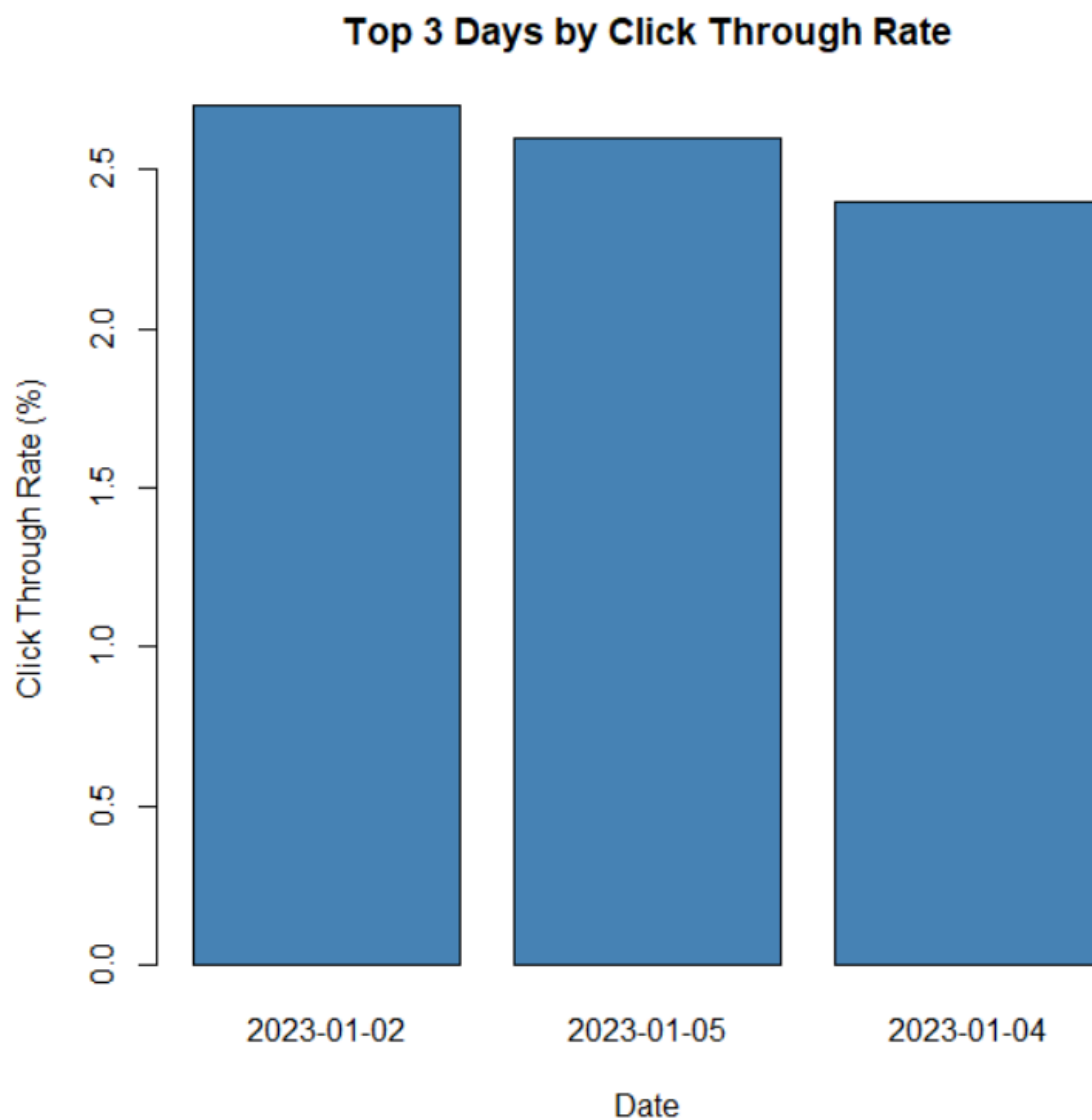
```
# Sort data by Click Through Rate
```

```
traffic <- traffic[order(-traffic$Click_Through_Rate), ]
```

```
# Select Top 3 Days
```

```
top3 <- head(traffic,3)
```

```
barplot(top3$Click_Through_Rate,  
        names.arg=top3$Date,  
        col="steelblue",  
        main="Top 3 Days by Click Through Rate",  
        xlab="Date",  
        ylab="Click Through Rate (%)")
```



>R (Stacked Area Chart)

```
#=====
```

```
# Question 3
```

```
# Stacked Area Chart
```

```
#=====
```

```
traffic <- read.csv("C:/Users/selco/OneDrive/Desktop/Website_Traffic.csv")
```

```
interaction <- t(as.matrix(traffic[,c("Likes","Shares","Comments")]))
```

```
colors <- c("skyblue","orange","lightgreen")
```

```
plot(1:5,
     traffic$Likes,
     type="n",
     ylim=c(0,max(colSums(interaction))),
```

```

xaxt="n",
xlab="Date",
ylab="User Interactions",
main="Distribution of User Interactions")

axis(1,
     at=1:5,
     labels=traffic$Date)

polygon(c(1:5,5:1),
        c(cumsum(interaction[1,]),rev(rep(0,5))),
        col=rgb(0.4,0.7,1,0.5),
        border=NA)

library(graphics)

matplot(t(interaction),
        type="l",
        lty=1,
        lwd=2,
        col=colors,
        xaxt="n",
        xlab="",
        ylab="")

legend("topleft",
       legend=c("Likes","Shares","Comments"),
       fill=colors)

```

